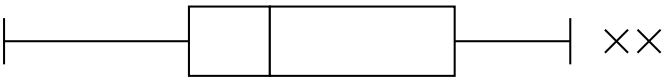
Representations Of Data

Solutions

This Chapter Overview

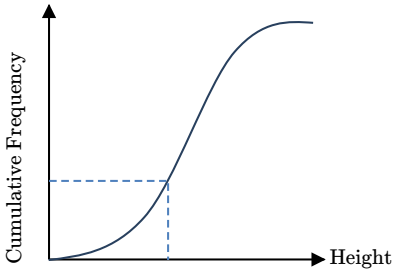
We've seen so far how data is collected and calculations can be made. We now concentrate on how the processed data can be *displayed*.

BOX PLOTS AND OUTLIERS

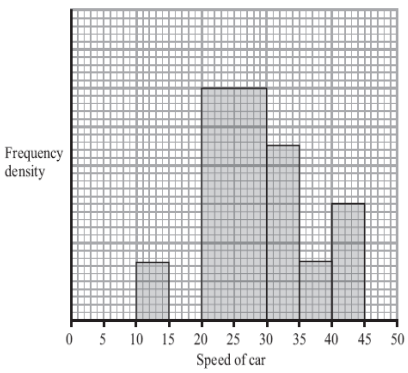


NEW since GCSE! Outliers.

CUMULATIVE FREQ DIAGRAMS



HISTOGRAMS



NEW since GCSE!
Area is not necessarily equal to frequency.
Forming a frequency polygon by joining midpoints.

Changes since the old S1 syllabus

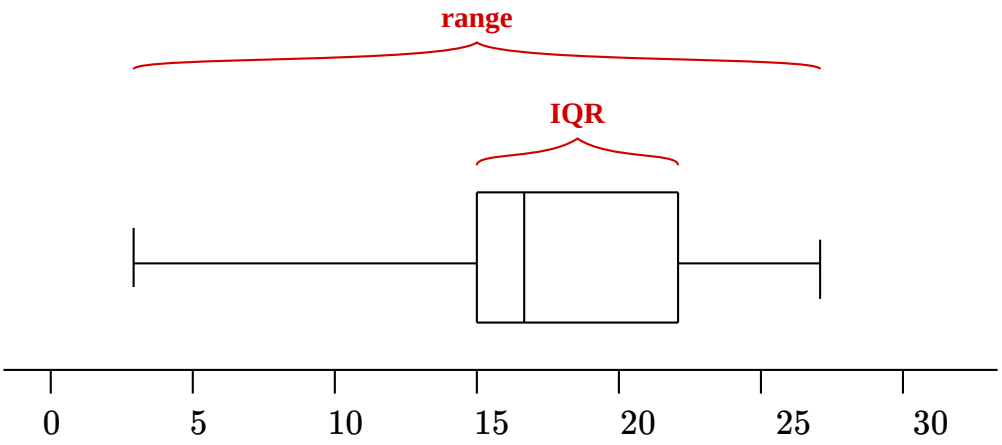
- Stem-and-leaf diagrams and skewness were removed.
- Cumulative frequency diagrams were added.
- Draw a frequency polygon from a histogram.

Box Plot recap

Solution 3

Box Plots allow us to visually represent the distribution of the data.

Minimum	Lower Quartile	Median	Upper Quartile	Maximum
3	15	17	22	27

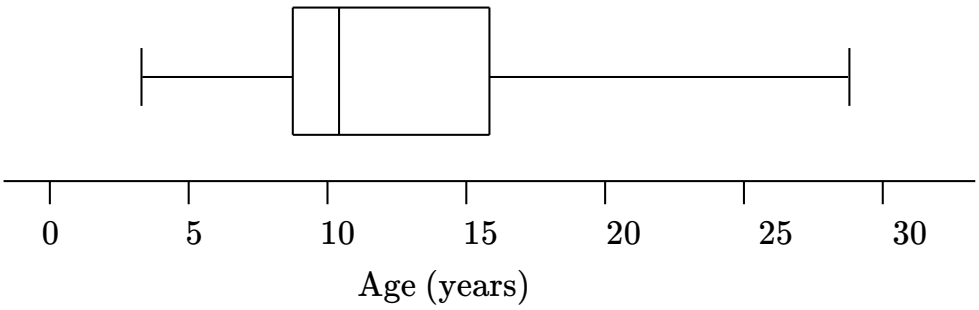


How is the **IQR** represented in this diagram?

How is the **range** represented in this diagram?

Interpreting a Box Plot

Solution 4



Decide whether each statement is true or false. Explain your reasoning.

"The right box represents more people than the left box."

False. Each box represents 25% of people, i.e. the same number of people!

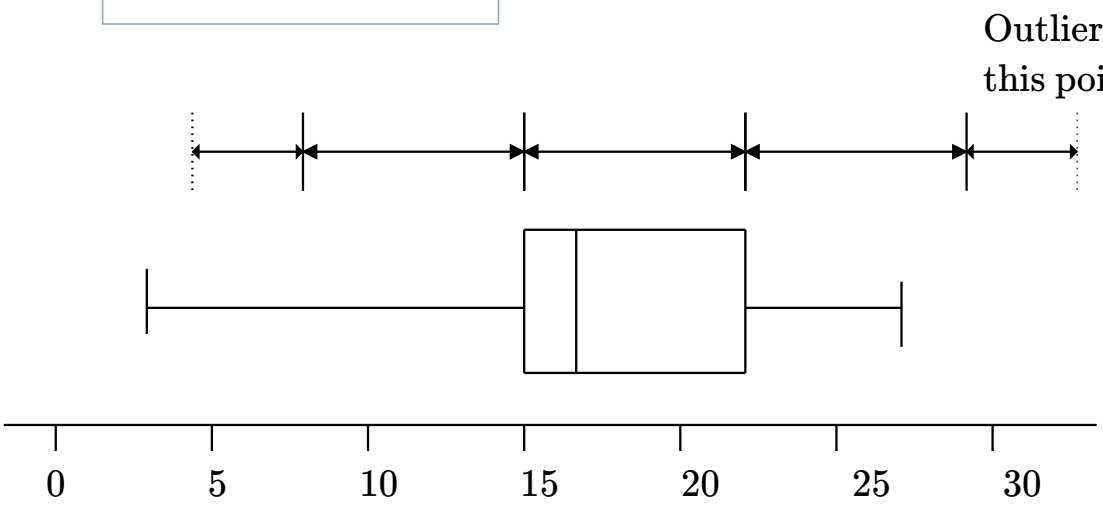
"The ages are more spread out above the median."

True. The wider the box or whisker, the more spread out the values are within that 25% of the data. We'd say that the data has "positive skew", but you are not required to know this term.

Outliers

Solution 5

An outlier is an extreme value.



One common definition of an outlier is when we're **1.5 IQRs** beyond the lower and upper quartiles.

(But you will be told in the exam if the rule differs from this)

Examples

Solution 6

The diameters of 11 different Roman coins are measured in centimetres:

2.2 2.5 2.7 2.7 2.8 3.0 3.1 3.1 3.2 4.0 4.7

Determine the quartiles and hence any outliers.

$Q_1 = 3^{rd}$ item $= 2.7$ **$Q_3 = 9^{th}$ item $= 3.2$**

$IQR = 3.2 - 2.7 = 0.5$

Lower outlier boundary $= 2.7 - 1.5 \times 0.5 = 1.95$

Upper outlier boundary $= 3.2 + 1.5 \times 0.5 = 3.95$

Therefore outliers are 4.0cm and 4.7cm.

[Textbook] The lengths, in cm, of 12 giant African land snails are given below:

17 18 18 19 20 20 20 20 21 23 24 32

a) Calculate the mean and standard deviation, given that $\Sigma x = 252$ and $\Sigma x^2 = 5468$.

b) An outlier is an observation which lies ± 2 standard deviations from the mean. Identify any outliers for this data.

a

$$\overline{x} = \frac{252}{12} = 21\text{cm}$$

$$\sigma = \sqrt{\frac{5468}{12} - 21^2} = 3.83(3sf)$$

b

Outlier boundaries: $21 - 2 \times 3.83 = 13.34$

$21 + 2 \times 3.83 = 28.66$

Outlier is just 32.

Context: Recall that the standard deviation is, roughly speaking, the average distance of each value from the mean. So the outlier definition is saying we're at least twice this average distance, which seems like a sensible definition.

In Year 2, you will encounter the **normal distribution**, which can be used to model data which is **clustered about some mean and tails off symmetrical in either direction**. If this data was approximately normally distributed, then there is a 5% chance a random observation would fall outside 2 standard deviations within the mean. You will learn then how to make such probability calculations.

Test Your Understanding — Solution 7

The ages of 15 Lib Dem MPs are given:

11 18 20 27 30 31 32 32 35 36 37 58 63 78 105

a) If an outlier is considered to be 1.5 interquartile ranges below the lower quartile or above the upper quartile, determine any outliers.

b) If instead an outlier is considered to be outside 2 standard deviations within the mean, determine any outliers. Note that $\Sigma x = 613$ and $\Sigma x^2 = 33815$

a

$$\mathbf{Q_1 = 27, \quad Q_3 = 58, \quad IQR = 31}$$

Boundaries: $\mathbf{27 - 1.5 \times 31 = -19.5}$

$\mathbf{58 + 1.5 \times 31 = 104.5}$

Therefore the 105 year old is an outlier.

b

$\mathbf{\overline{x} = \frac{613}{15} = 40.9}$

$\mathbf{\sigma = \sqrt{\frac{33815}{15} - \left(\frac{613}{15}\right)^2} = 24.2 \ \ (3sf)}$

Boundaries: $\mathbf{40.9 + 24.2 \times 2 = 89.3}$

$\mathbf{40.9 - 24.2 \times 2 = -7.5}$

Therefore the 105 year old is an outlier.

Box Plot Example — Solution 8

Smallest values	Largest values	Lower Quartile	Median	Upper Quartile
0, 3	21, 27	8	10	14

Draw a box plot to represent the above data.

$$IQR = 14 - 8 = 6$$

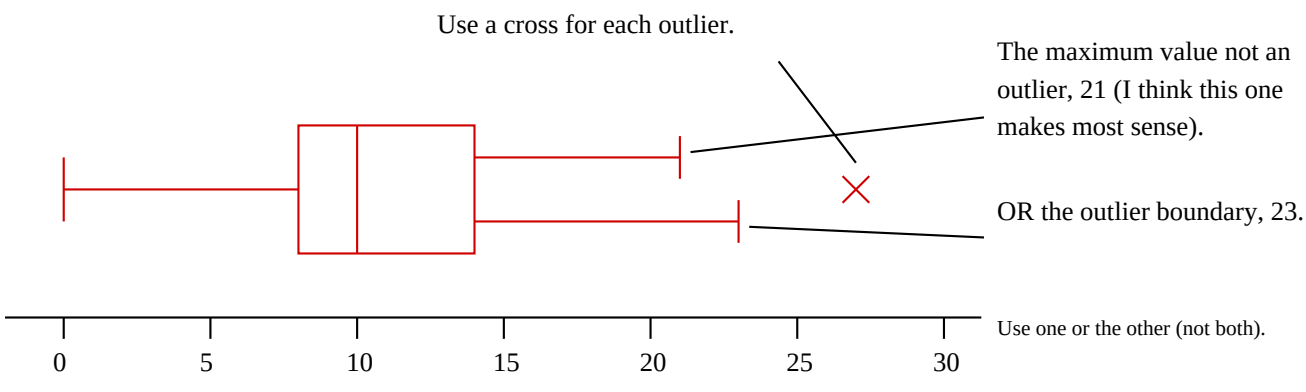
Outlier boundaries:

$\mathbf{14 + (1.5 \times 6) = 23}$

$\mathbf{8 - (1.5 \times 6) = -1}$

Exam Tip: You MUST show your outlier boundary calculations.

When there is an outlier at one end, there are two allowable places to put the end of the whisker:



Test Your Understanding

Solution 9

Jan 2011 Q3. A small company recorded monthly sales over a long period. An outlier is more than 1.5 interquartile ranges beyond either quartile.

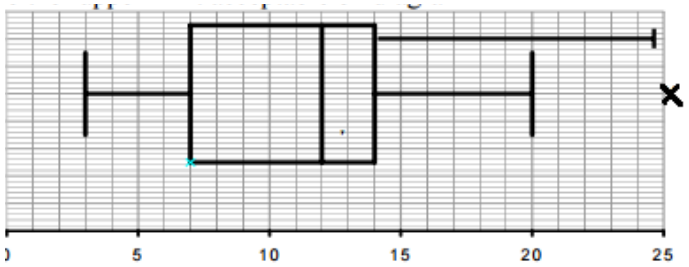
(a) Draw a box plot on graph paper indicating outliers. (5 marks)

(c) The company claims that sales exceed £10,000 in 75% of months. Comment on this claim. (2 marks)

Summary	Sales (£1000s)
Two lowest	3, 4
Lower quartile	7
Median	12
Upper quartile	14
Two highest	20, 25

a

$14 + 1.5(14 - 7) = 24.5; 7 - 1.5(14 - 7) = -3.5$. The outlier is 25. The upper whisker may stop at 20 or the upper fence.



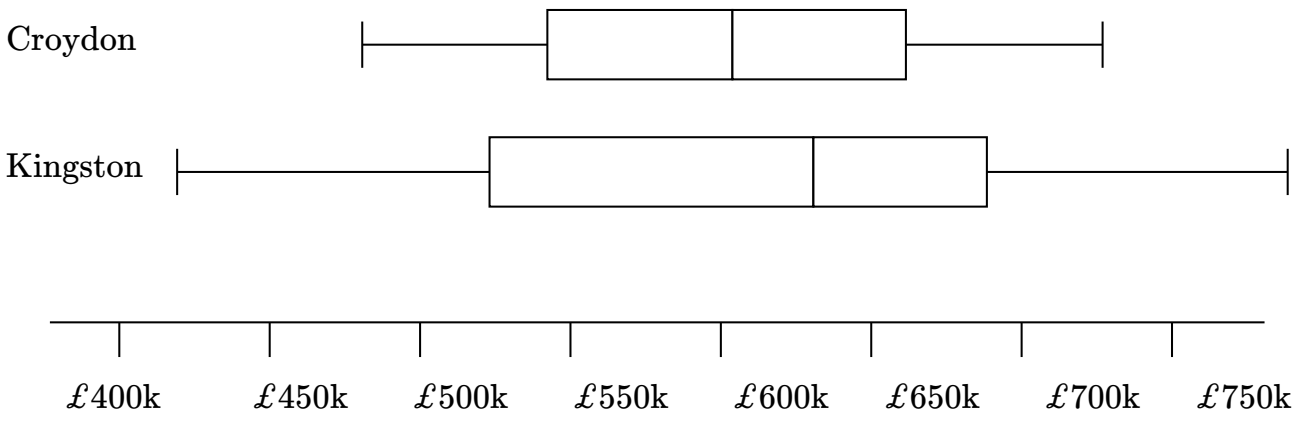
c

The claim is unsupported. The lower quartile is £7000, so 75% of months have at least £7000; £10,000 lies inside the box.

Comparing Box Plots

Solution 10

Box Plot comparing house prices of Croydon and Kingston-upon-Thames:



"Compare the prices of houses in Croydon with those in Kingston". (2 marks)

For 1 mark, one of:

- The interquartile range of house prices in Kingston is greater than Croydon.
- The range of house prices in Kingston is greater than Croydon.

Include some measure of spread.

For 1 mark:

"The **median** house price in Kingston was greater than that in Croydon."

Include some measure of **location** (median is best).

Exercise 3A – Statistics Year 1 Pages 43

Solution 11

4. The ages of nine people at a children's party have $\sum x = 92, \sum x^2 = 1428$.

(a) Calculate the mean and standard deviation. (3 marks)

An outlier lies more than two standard deviations from the mean. One age is 30.

(b) Is 30 an outlier? Give a reason. (2 marks)

(c) Suggest why it could be legitimate. (1 mark)

(d) Given that all nine were children, clean the data and recalculate the mean and standard deviation. (3 marks)

Hint: find new values of n, $\sum x$ and $\sum x^2$.

a

$\bar{x} = 92/9 = 10.2.$

$\sigma = \sqrt{1428/9 - (92/9)^2} = 7.36$ (3 s.f.).

b

The limits are $\bar{x} \pm 2\sigma = -4.50, 24.9$. Since $30 > 24.9$, it is an outlier.

c

It could be a parent's age.

d

Remove 30. $\sum x = 92 - 30 = 62, \sum x^2 = 1428 - 900 = 528.$

$\bar{x} = 62/8 = 7.75, \sigma = \sqrt{528/8 - (62/8)^2} = 2.44$.

Exercise 3B – Statistics Year 1 Pages 45

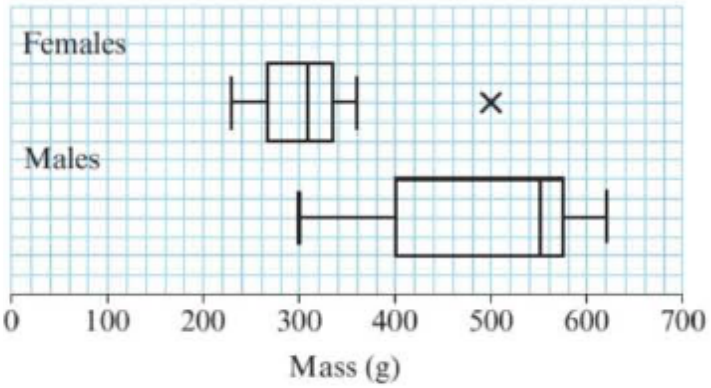
Solution 12

3. The box plots show male and female turtle masses in grams.

(a) Compare their masses.

(b) A turtle weighs 330 g. Is it more likely male or female? Give a reason.

(c) Write down the mass of the largest female.



a

Males have a higher median mass and a larger range and interquartile range.

b

More likely female. Few males are this light; roughly three quarters of females weigh less.

c

500 g, even though it is an outlier.

Exercise 3B – Statistics Year 1 Pages 45

Solution 13

4. The table gives daily maximum gusts (knots) in Camborne in September 1987 from the large data set. © Crown Copyright Met Office.

(a) Calculate Q_1, Q_2, Q_3. (3 marks)

(b) Using fences 1.5 IQRs beyond the quartiles, show that 46 and 78 are outliers. (1 mark)

(c) Draw a box plot. (3 marks)

13	17	19	20	21
21	22	23	24	25
25	25	26	26	26
27	29	30	30	30
33	35	38	46	78

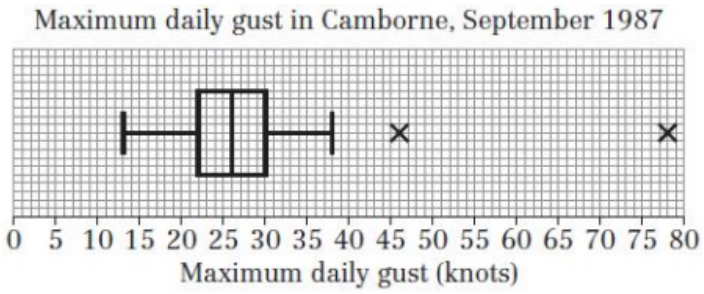
a

$25/4 = 6.25$: use the 7th value, 22. The median is the 13th value, 26. $3(25)/4 = 18.75$: use the 19th value, 30.

b

$\text{IQR} = 30 - 22 = 8$. The fences are $22 - 1.5(8) = 10$ and $30 + 1.5(8) = 42$. Both 46 and 78 exceed 42.

c



Supplementary Questions:

Solution 14

5. May 2006 Q1. Children from schools A and B run for charity. Times are recorded to the nearest minute. Figure 1 summarises school A.

(a) Describe the features and uses of a box plot. (3 marks)

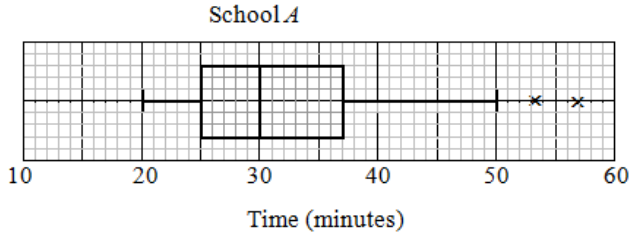
(b) (i) By what time had 75% of A finished? (ii) Name this value. (2 marks)

(c) Explain the two crosses. (2 marks)

School B had minimum 25, quartiles 30, 37 and 50, and maximum 55 minutes.

(d) Draw a box plot for B. (4 marks)

(e) Compare the two plots. (4 marks)



a

A box plot shows quartiles, median and extremes; it identifies spread, skewness and outliers and permits comparisons. Any three features or uses.

b

(i) 37 minutes. (ii) Upper quartile, or 75th percentile.

c

The crosses are outliers. These children took unusually long and may have walked. Treat their times with caution.

Solutions to (d) and (e) follow.

Supplementary Questions: Solution 15

5. May 2006 Q1. Children from schools A and B run for charity. Times are recorded to the nearest minute. Figure 1 summarises school A.

(a) Describe the features and uses of a box plot. (3 marks)

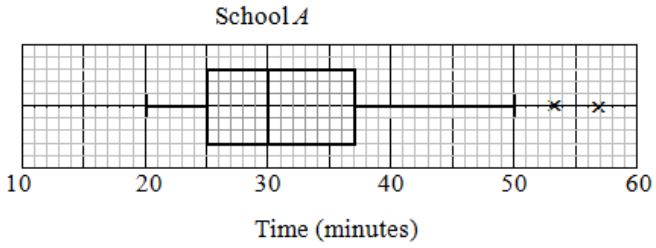
(b) (i) By what time had 75% of A finished? (ii) Name this value. (2 marks)

(c) Explain the two crosses. (2 marks)

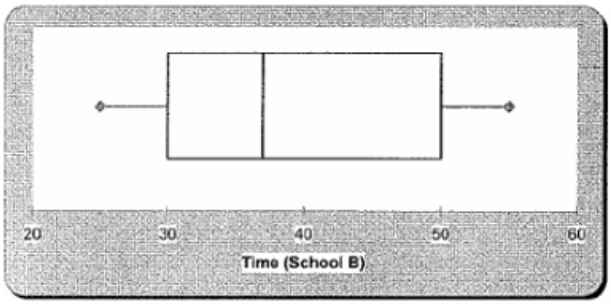
School B had minimum 25, quartiles 30, 37 and 50, and maximum 55 minutes.

(d) Draw a box plot for B. (4 marks)

(e) Compare the two plots. (4 marks)



d



e

Children in A generally took less time. For example, 75% of A took at most 37 minutes, compared with 50% of B. The quartiles and median are lower in A. A has outliers and B has none. Both are positively skewed. A has a smaller IQR but a larger range. Any four valid comparisons.

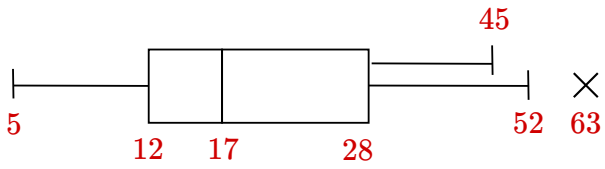
Supplementary Questions: Solution 16

6. June 2005 Q4. Take-off delays from City A to B had minimum 5 and maximum 63 minutes. A quarter were at most 12 minutes, half at most 17 and three quarters at most 28. Only one exceeded 45 minutes. Use fences 1.5 IQRs beyond the quartiles.

(a) Draw a box plot. (7 marks) (b) Comment on the distribution and justify. (2 marks) (c) Interpret it for a frequent passenger. (1 mark)

a

$1.5(28-12)=24$. Fences: $12-24=-12$ and $28+24=52$. Only 63 is an outlier.



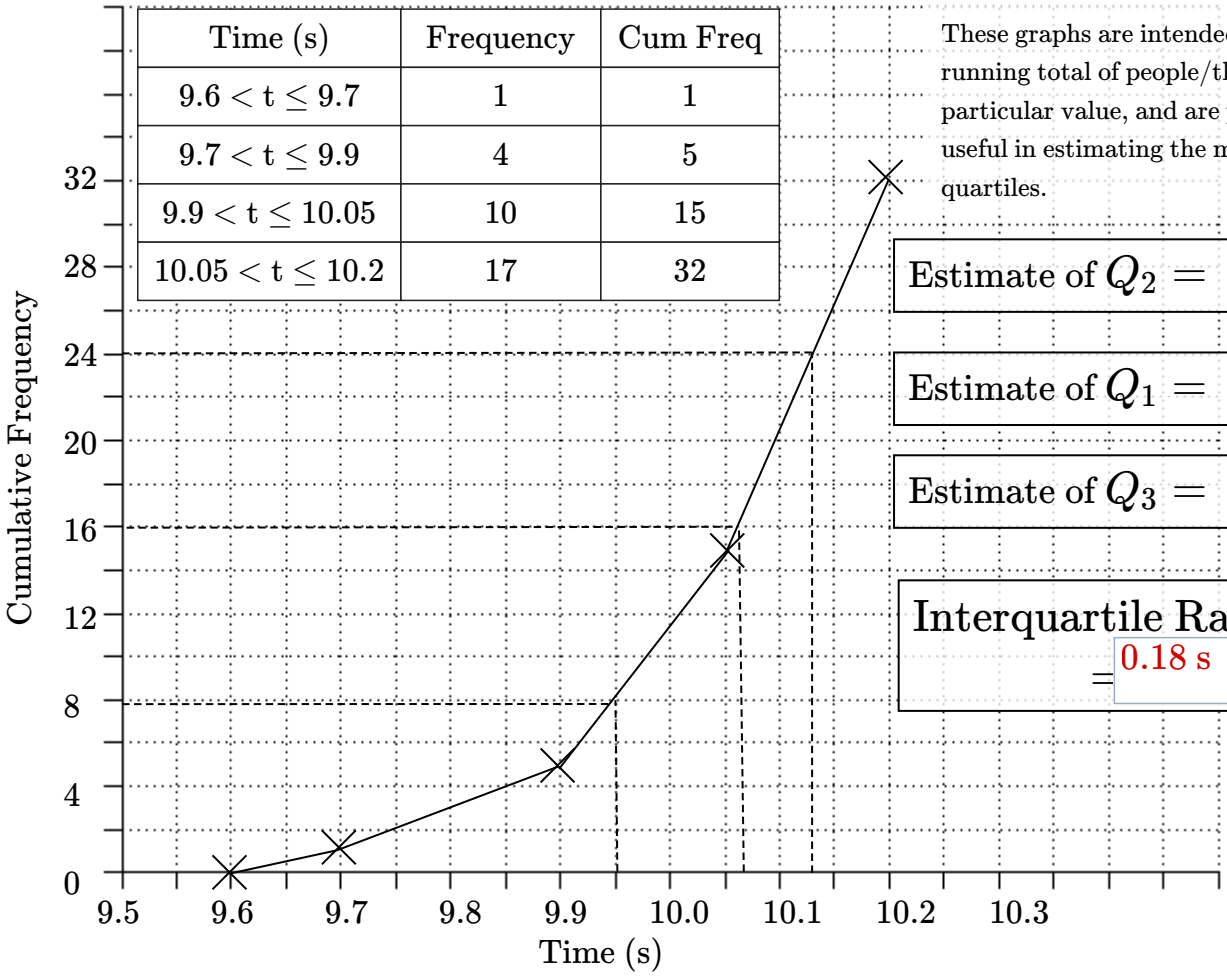
b Positive skew: $17-12=5<28-17=11$.

c Many delays are short, which passengers may find acceptable.

Cumulative Frequency Diagrams

Solution 17

Time (s)	Frequency	Cum Freq
$9.6 < t \leq 9.7$	1	1
$9.7 < t \leq 9.9$	4	5
$9.9 < t \leq 10.05$	10	15
$10.05 < t \leq 10.2$	17	32



These graphs are intended to show the running total of people/things **up to** a particular value, and are particularly useful in estimating the median and quartiles.

Estimate of Q_2 = 10.07 s

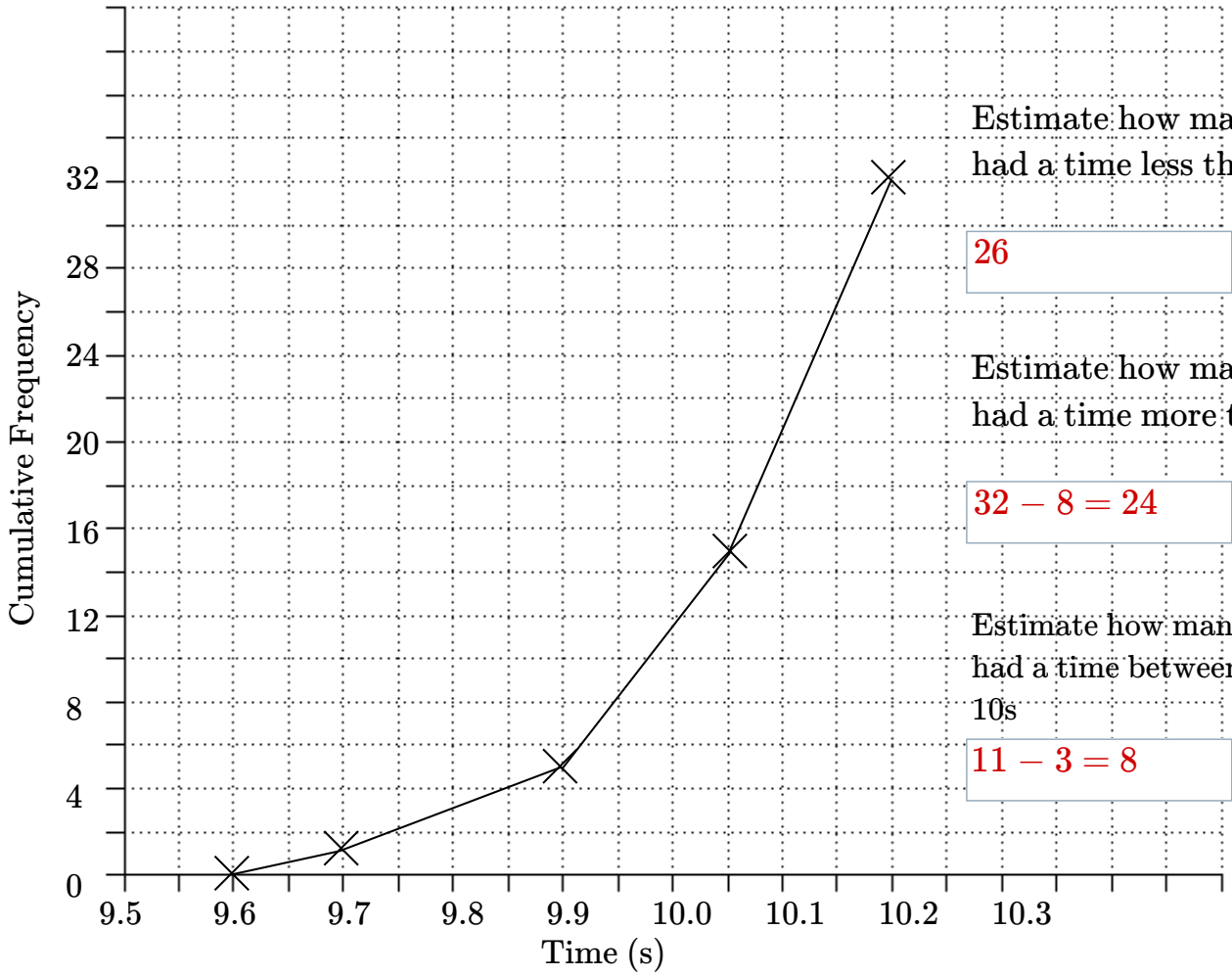
Estimate of Q_1 = 9.95 s

Estimate of Q_3 = 10.13 s

Interquartile Range = 0.18 s

Cumulative Frequency Diagrams

Solution 18



Estimate how many runners had a time less than 10.15s.

26 runners

Estimate how many runners had a time more than 9.95

$32 - 8 = 24$ runners

Estimate how many runners had a time between 9.8s and 10s

$11 - 3 = 8$ runners

Exercise 3C – Statistics Year 1 Pages 47-48

Solution 19

4. A vet records male and female domestic shorthair cat masses in kg.

(a) Draw cumulative frequency diagrams on common axes.

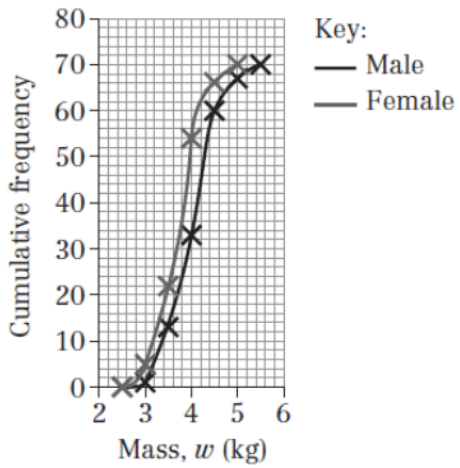
(b) Which group has greater spread?

A female is underweight below 3.2 kg; a male is underweight below 3.8 kg.

(c) Which group has fewer underweight cats?

Mass (kg)	Male	Female
$2.5 \leq w < 3$	1	5
$3 \leq w < 3.5$	12	17
$3.5 \leq w < 4$	20	32
$4 \leq w < 4.5$	27	12
$4.5 \leq w < 5$	7	4
$5 \leq w < 5.5$	3	0

a



b

Male IQR $= 4.4 - 3.6 = 0.8$ kg. Female IQR $= 4.0 - 3.4 = 0.6$ kg. Males have greater spread.

c

About 12 females and 25 males are underweight. There are fewer underweight females.

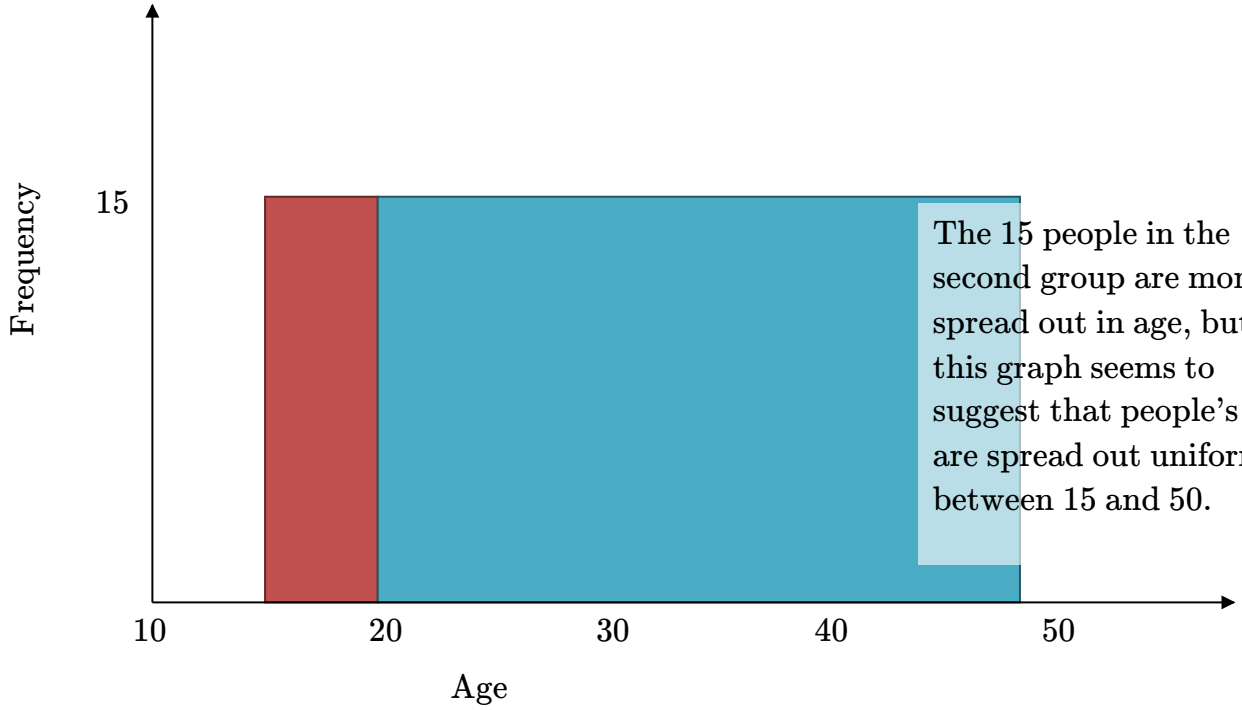
Histograms

Solution 20

Age (years)	Frequency
$15 \leq a < 20$	15
$20 \leq a < 50$	15

Pablo is hosting a party. He counts how many people are between 15 and 20, and 20 and 50.

Why is below graph somewhat unhelpful. How could we fix it?



The 15 people in the second group are more spread out in age, but this graph seems to suggest that people's ages are spread out uniformly between 15 and 50.

Estimating frequency within an age group

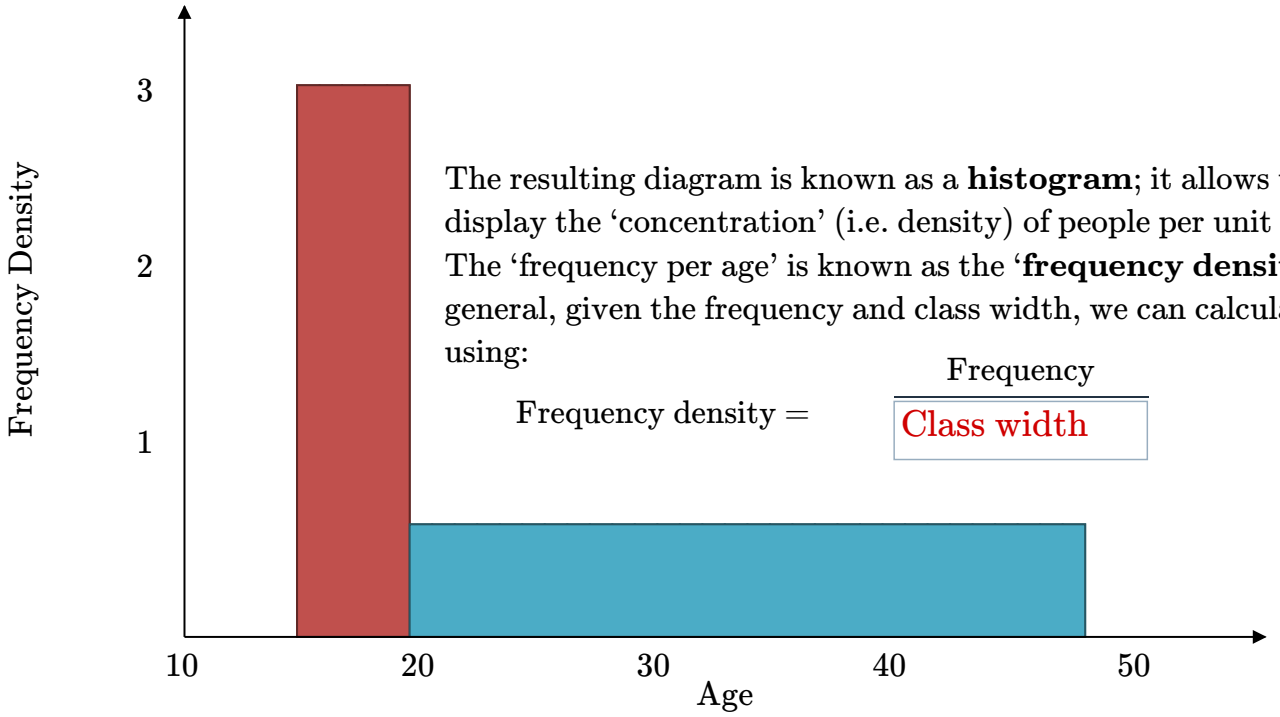
Solution 21

Age (years)	Frequency
$15 \leq a < 20$	15
$20 \leq a < 50$	15

Assume ages are evenly spread within each age group.

3 people of each age in the 15–20 group.

0.5 people of each age in the 20–50 group.



The resulting diagram is known as a **histogram**; it allows us to display the 'concentration' (i.e. density) of people per unit value. The 'frequency per age' is known as the '**frequency density**'. In general, given the frequency and class width, we can calculate it using:

$$\text{Frequency density} = \frac{\text{Frequency}}{\text{Class width}}$$

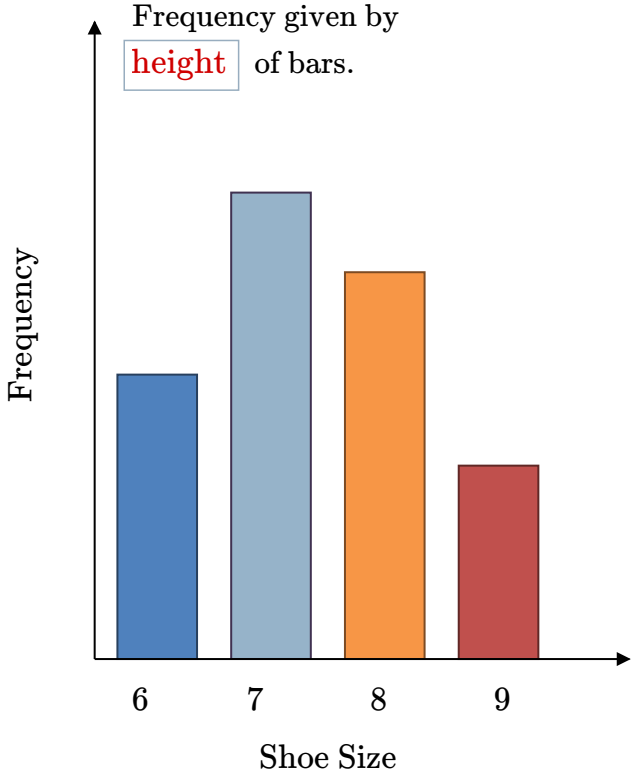
Bar Charts vs Histograms

Solution 22

Bar charts

For discrete data.

Frequency given by height of bars.



Histograms

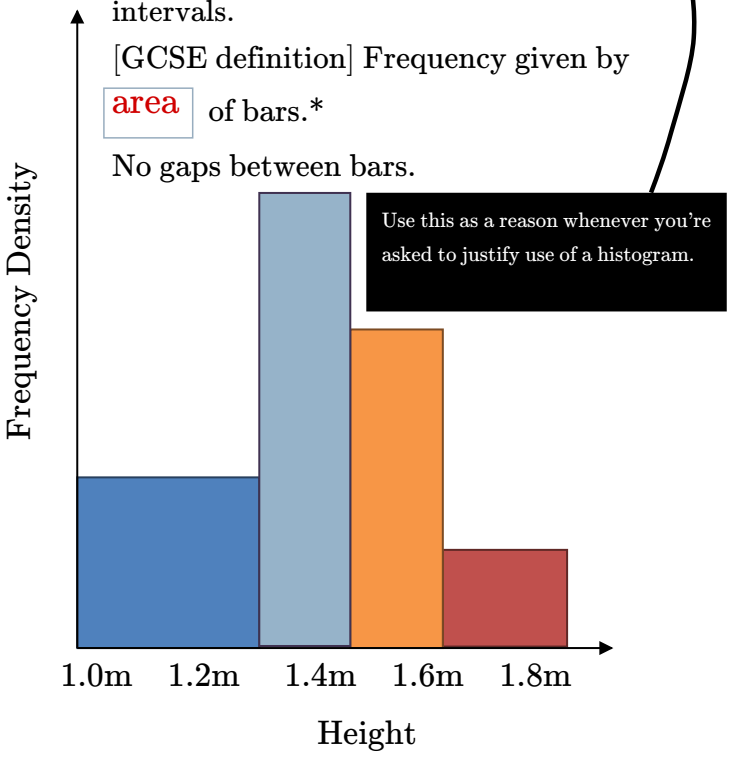
For continuous data.

Data divided into (potentially uneven) intervals.

[GCSE definition] Frequency given by area of bars.*

No gaps between bars.

Use this as a reason whenever you're asked to justify use of a histogram.



* Not necessarily true. We'll correct this in a sec.

Bar Charts vs Histograms

Solution 23

Q1

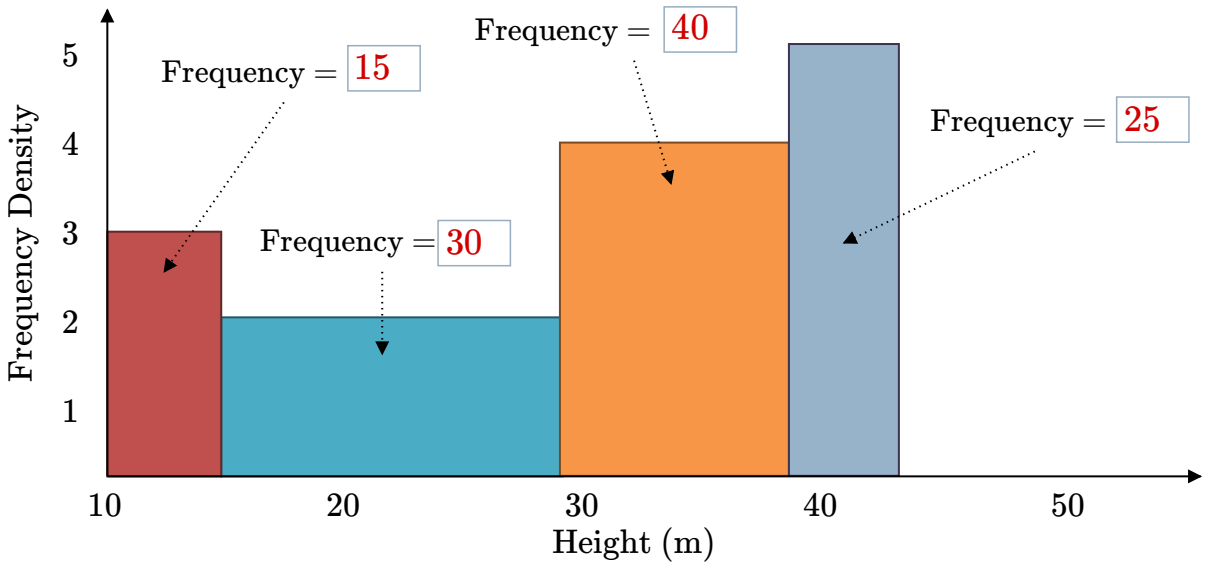
Weight (w kg)	Frequency	Frequency Density
$0 < w \leq 10$	40	4
$10 < w \leq 15$	6	1.2
$15 < w \leq 35$	52	2.6
$35 < w \leq 45$	10	1

Still using the **incorrect** GCSE formula:

Freq

F.D. Width

Q2



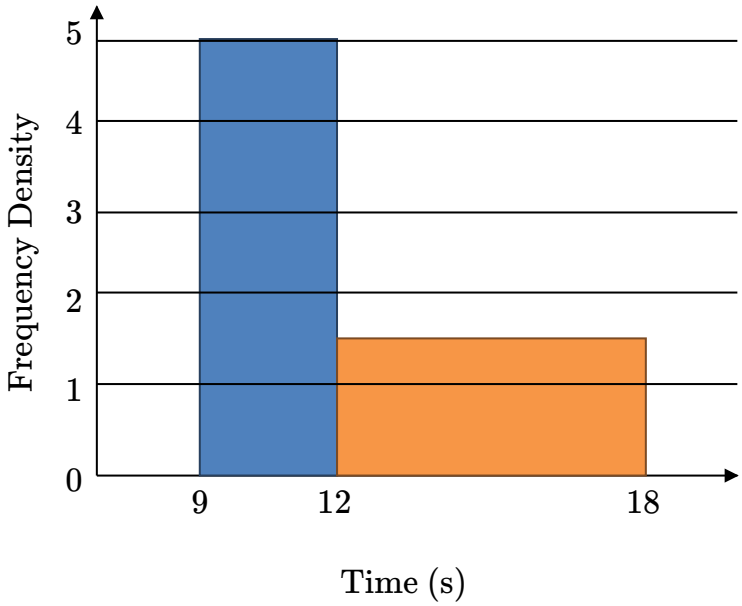
SKILL #1: Area = frequency?

Solution 24

Unlike at GCSE, the area of a bar is not necessarily equal to the frequency; there are just **proportional**.

✎ Identify the scaling $area \times k = frequency$ using a known area with known frequency (which may be total area/frequency or just one bar)

There were 60 runners in a 100m race. The following histogram represents their times. Determine the number of runners with times above 14s.



Total frequency is known; therefore find total area and hence the 'scaling'.

Total area $= 15 + 9 = 24$

Area		Freq	
24	$\times k$	60	$k = \frac{60}{24} = 2.5$

Then use this scaling along with the desired area.

Area $= 4 \times 1.5 = 6$

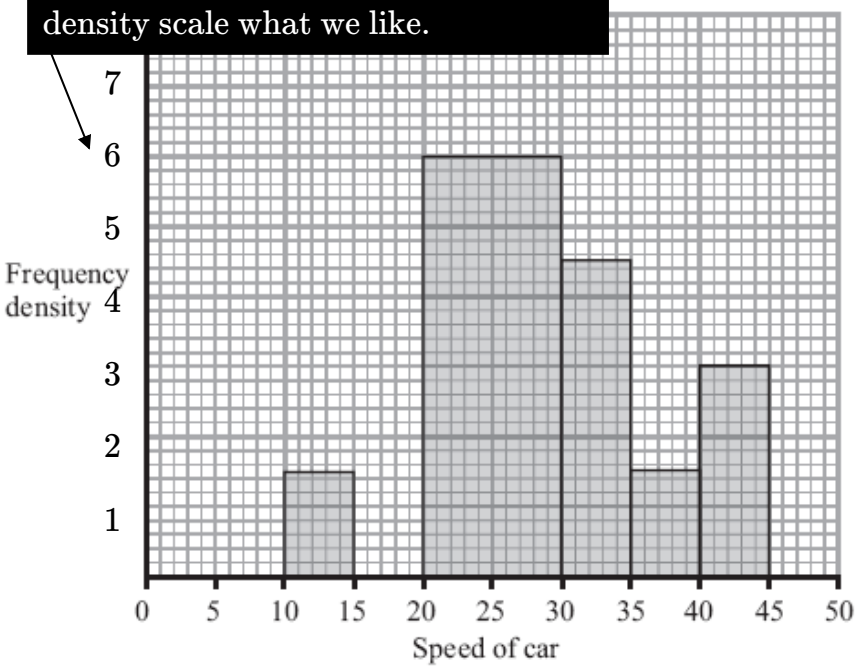
Frequency $= 6 \times 2.5 = 15$

Edexcel S1 May 2012 Q5

A policeman records the speed of the traffic on a busy road with a 30 mph speed limit. He records the speeds of a sample of 450 cars. The histogram in Figure 2 represents the results.

(a) Calculate the number of cars that were exceeding the speed limit by at least 5 mph in the sample. *(4 marks)*

Tip: We can make the frequency density scale what we like.



M1 A1: Determine what one small square or one large square is worth.

(i.e. work out $area \rightarrow freq$ scaling)

Area Freq

$112.5 \times k 450$

$k = \frac{450}{112.5} = 4$

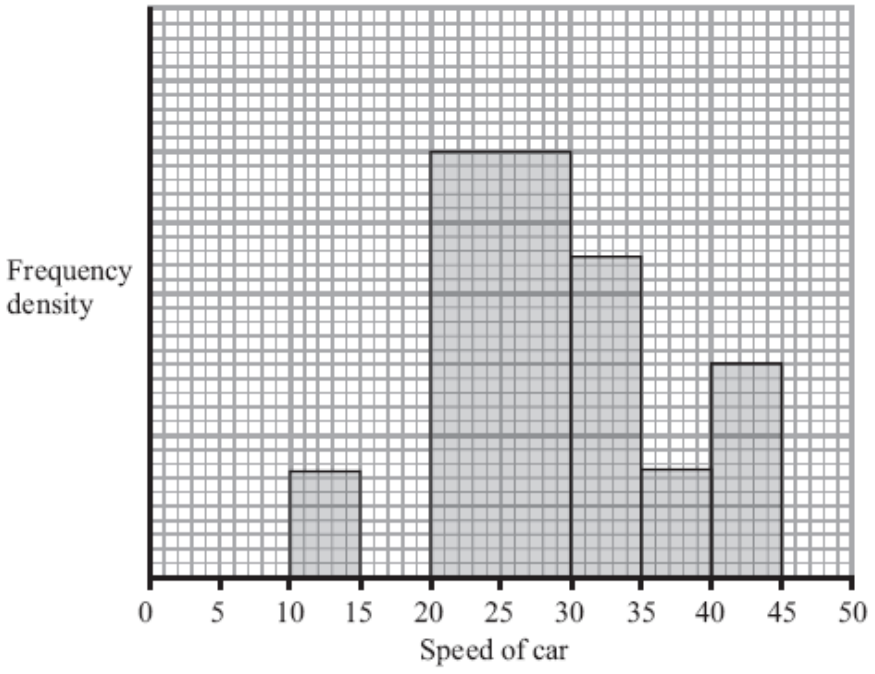
M1 A1: Use this to find number of cars travelling >35mph.

Area Freq

$22.5 \times 4 \quad 90$

Write: $22.5 \times 4 = 90$

(b) Estimate the value of the mean speed of the cars in the sample. *(3 marks)*



M1 M1: Use histogram to construct sum of speeds.

$\frac{30 \times 12.5 + 240 \times 25 + \ldots}{450}$

A1 Correct value

$= 28.8$

Tip: Whenever you are asked to calculate mean, median or quartiles from a histogram, form a grouped frequency table. Use your scaling factor to work out the frequency of each bar.

(*c*) Estimate, to 1 decimal place, the value of the median speed of the cars in the sample. **(2)**
(*d*) Comment on the shape of the distribution. Give a reason for your answer. **(2)**
(*e*) State, with a reason, whether the estimate of the mean or the median is a better representation of the average speed of the traffic on the road. **(2)**

(crossed out questions would not appear in new syllabus)

c

$Q_2 = 20 + (195/240)(10) = 28.125 \approx 28.1$ mph. Using position 195.5 gives 28.145 mph.

d

$Q_2 < \bar{x}$, indicating positive skew. An almost symmetric interpretation is also acceptable if the rounded estimates are treated as close.

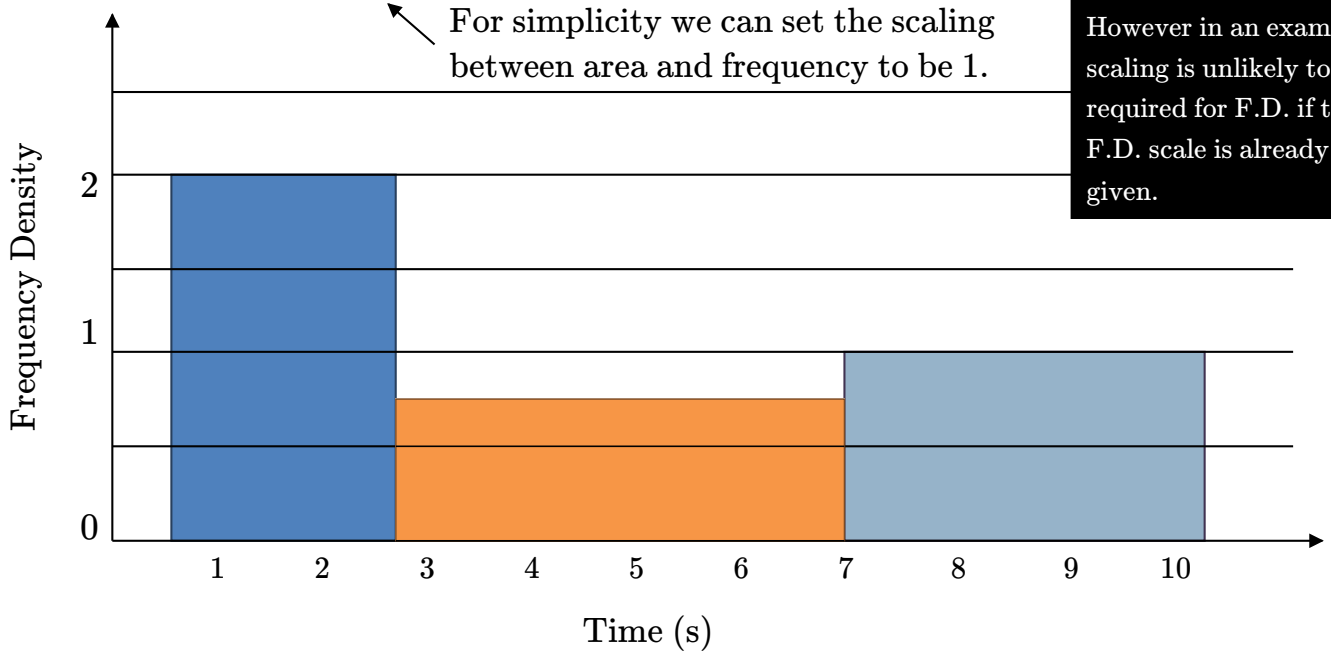
e

For a skewed distribution choose the median: it is robust to extreme values. For an approximately symmetric distribution the mean uses all the data.

Weight (to nearest kg)	Frequency	F.D.
1-2	4	$4 \div 2 = 2$
3-6	3	$3 \div 4 = 0.75$
7-9	$3 \times 1 = 3$	1

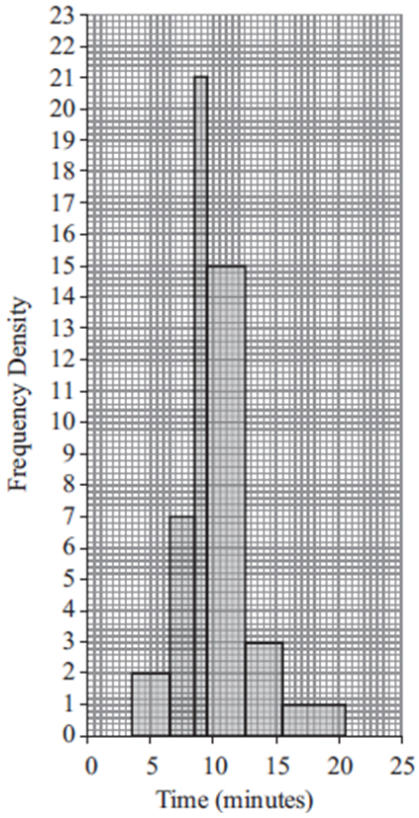
For simplicity we can set the scaling between area and frequency to be 1.

Note the gaps affects class width!
Remember the frequency density axis is only correct to scale, so there may be some scaling. However in an exam scaling is unlikely to be required for F.D. if the F.D. scale is already given.



Test Your Understanding — Solution 29

Jan 2012 Q1. The histogram shows the time, to the nearest minute, that a random sample of 100 motorists were delayed by roadworks on a stretch of motorway.



(a) Complete the table. (2 marks)

Delay (minutes)	Number of motorists
4–6	6
7–8	14
9	21
10–12	45
13–15	9
16–20	5

Tip: Use the correct class widths.

(b) Estimate the number of motorists delayed between 8.5 and 13.5 minutes by the roadworks. (2 marks)

b $21 + 45 + 3 = 69$

SKILL #3: Width and height on diagram — Solution 30

An exam favourite is to ask what width and height we'd draw a bar in a drawn histogram.

Q: The frequency table shows some running times. On a histogram the bar for 0-4 seconds is drawn with width 6cm and height 8cm. Find the width and height of the bar for 4-6 seconds.

Time (seconds)	Frequency
$0 \leq t < 4$	8
$4 \leq t < 6$	9

Strategy **Tip**: Find the scaling for class width to drawn width and frequency density to drawn height.

Solution

For 0-4 bar:

Class width $= 4$

Frequency density $= 8 \div 4 = 2$

$\therefore$ Scaling for width: 1.5

Scaling for height: 4

4-6 bar: class width 2, frequency density 4.5

$Width = 2 \times 1.5 = 3cm$

$Height = 4.5 \times 4 = 18cm$

Test Your Understanding

Solution 31

May 2009 Q3. x was measured to the nearest whole number. Forty observations are shown. A histogram bar for 10–15 is 2 cm wide and 5 cm high.
Find (a) the width (1 mark) and (b) the height (2 marks) of the bar for 16–18.

x	10–15	16–18	19–
Frequency	15	9	16

a

The class widths are 6 and 3, so the second bar is $2(3/6) = 1$ cm wide.

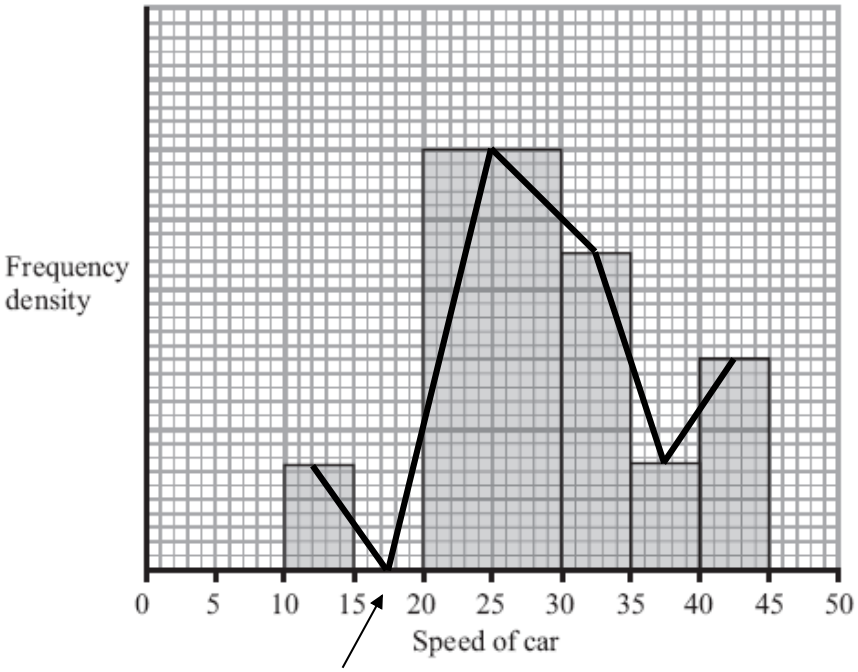
b

$10\ \text{cm}^2$ represents 15 observations. For frequency 9 the area is $9(10/15) = 6\ \text{cm}^2$. With width 1 cm the height is 6 cm.

SKILL #4 :: Forming a frequency polygon

Solution 32

Recall that a frequency polygon can be drawn by using the midpoint of each interval. This corresponds to the midpoint of the top of each bar in a histogram.

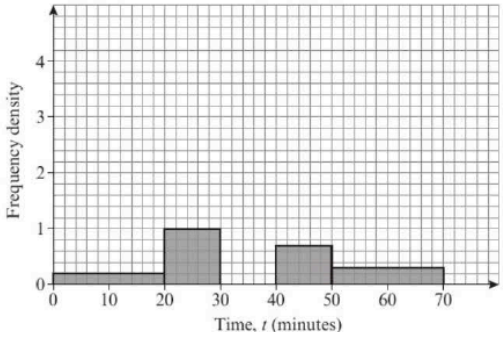


Note that the frequency in this interval is 0. That needs to be reflected in the frequency polygon.

Exercise 3D – Statistics Year 1 Pages 50-52 Solution 33

5. The partly completed histogram shows airport passenger delays.
(a)(i) Copy and complete the table. (ii) Copy and complete the histogram. (4 marks)
(b) Estimate how many passengers were delayed between 25 and 38 minutes. (2 marks)

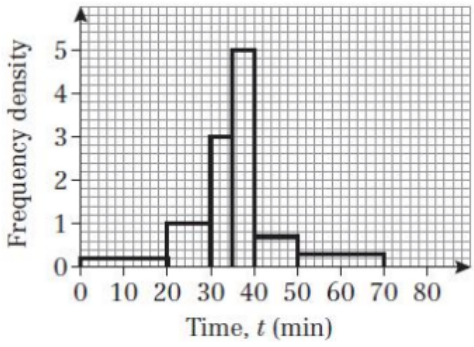
Time (min)	Frequency
$0 \le t < 20$	4
$20 \le t < 30$	
$30 \le t < 35$	15
$35 \le t < 40$	25
$40 \le t < 50$	
$50 \le t < 70$	



a(i)

Time (min)	Frequency	Width	Density
$0 \le t < 20$	4	20	0.2
$20 \le t < 30$	10	10	1
$30 \le t < 35$	15	5	3
$35 \le t < 40$	25	5	5
$40 \le t < 50$	7	10	0.7
$50 \le t < 70$	6	20	0.3

a(ii)



b

$\frac{5}{10}(10) + 15 + \frac{3}{5}(25) = 35$ passengers.

Exercise 3D – Statistics Year 1 Pages 50-52 Solution 34

6. The variable y was measured to the nearest whole number. 60 observations were taken and are recorded in the table below.

y	10–12	13–14	15–17	18–25
Frequency	6	24	18	12

(a) Write down the class boundaries for the 13–14 class. (1 mark)

A histogram was drawn and the bar representing the 13–14 class had a width of 4 cm and a height of 6 cm. For the bar representing the 15–17 class, find:
(i) the width (1 mark)
(ii) the height (2 marks).

Problem-solving: Remember that area is proportional to frequency.

a

12.5 and 14.5 are the boundaries because the data are continuous.

b(i)

The boundaries are 14.5 and 17.5. The width is $17.5 - 14.5 = 3 = 1.5 \times 2$.
The bar width is $1.5 \times 4 = 6$ cm.

b(ii)

The 13–14 frequency density is $24/2 = 12$. The 15–17 density is $18/3 = 6$, half the first density.
The bar height is $0.5 \times 6 = 3$ cm.

Exercise 3D – Statistics Year 1 Pages 50-52

Solution 35

7. Daily mean temperatures in Leeming in May 2015 are summarised below (large data set; © Crown Copyright Met Office). The $8 \le t < 10$ bar is 1 cm wide and 8 cm high.

(a) Find the width and height of the $11 \le t < 12$ bar. (2 marks)
(b) Estimate the mean and standard deviation. (3 marks)
(c) Interpolate the lower quartile. (2 marks)
(d) Estimate the number of days exceeding the mean plus one standard deviation. (2 marks)

Temperature (°C)	Frequency
$4 \le t < 8$	4
$8 \le t < 10$	8
$10 \le t < 11$	6
$11 \le t < 12$	7
$12 \le t < 15$	5
$15 \le t < 16$	1

a

Width $= 0.5$ cm. Its density is 7, and height is twice density: 14 cm.

b

Using midpoints, $\bar{t} = 322.5/31 = 10.4°\text{C}$; $\sigma = \sqrt{113.89 - (322.5/31)^2} = 2.4°\text{C}$.

c

Position $31/4 = 7.75$. $Q_1 = 8 + 2(7.75 - 4)/8 = 8.94°\text{C}$.

d

Mean plus SD is $12.8°\text{C}$. Up to 12.8 there are approximately $25 + 5(12.8 - 12)/3 = 26.33$ days; $31 - 26.33 \approx 4.7$ days exceed it.

Exercise 3E – Statistics Year 1 Pages 54

Solution 36

4. The table gives sampled daily maximum relative humidities from the large data set for Leuchars and Camborne during 2015. © Crown Copyright Met Office.

Leuchars	100	98	100	100	100	100	100	100	94	100	91	100	100	89	100
Camborne	92	95	99	96	100	100	90	98	81	99	100	99	91	98	100

(a) Find the median and quartiles for both samples. (4 marks)
(b) Compare the samples. (2 marks)

a

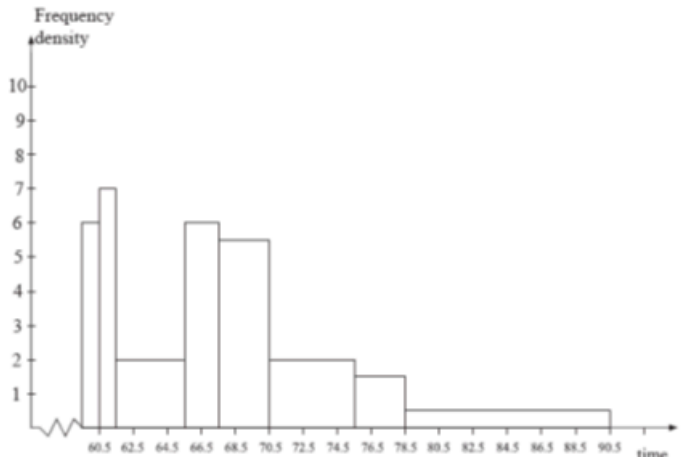
Sort each sample and use positions 4, 8 and 12.
Leuchars: $Q_1 = 98$, median $= 100$, $Q_3 = 100$.
Camborne: $Q_1 = 92$, median $= 98$, $Q_3 = 100$.

b

Leuchars has the higher median humidity. Camborne has the greater spread: its IQR is $100 - 92 = 8$, compared with $100 - 98 = 2$ in Leuchars.

Supplementary Exercise

Jan 2008 Q3. The histogram shows time, measured to the nearest minute, for 140 runners to complete a fun run. Calculate how many took between 78.5 and 90.5 minutes. (5 marks)



Solution 37

3

Total bar area $= 1(6) + 1(7) + 4(2) + 2(6) + 3(5.5) + 5(2) + 3(1.5) + 12(0.5) = 70.$

The final bar has area $(90.5 - 78.5)(0.5) = 6.$

Number of runners $= 6(140/70) = 12.$

Width	1	1	4	2	3	5	3	12
Density	6	7	2	6	5.5	2	1.5	0.5

Supplementary Exercise

2. The distances, to the nearest km, travelled by 134 examiners to a London meeting are summarised below.

(a) Justify the use of a histogram. (1 mark)

(b) Calculate the frequency densities needed. Do not draw the histogram. (2 marks)

Distance (km)	Examiners
41–45	4
46–50	19
51–60	53
61–70	37
71–90	15
91–150	6

Solution 38

a

Distance is continuous; a histogram represents grouped continuous data with unequal class widths.

b

Use the class boundaries, allowing for rounding to the nearest kilometre. Divide frequency by width:

$4/5 = 0.8, 19/5 = 3.8, 53/10 = 5.3, 37/10 = 3.7, 15/20 = 0.75, 6/60 = 0.1.$

Supplementary Exercise

May 2013 (R) Q3. Yields of 70 tomato plants are given in the table. Use $\sum fx = 755$, $\sum fx^2 = 12037.5$. The $5 \le y < 10$ bar is 1.5 cm wide, 8 cm high.

(a) Find the width and height for $15 \le y < 25$. (3 marks)
(b) Interpolate the median. (2 marks)
(c) Estimate the mean and standard deviation. (4 marks)
(d) Describe and justify skewness. (2 marks)

Yield (kg)	Frequency	Midpoint
$0 \le y < 5$	16	2.5
$5 \le y < 10$	24	7.5
$10 \le y < 15$	14	12.5
$15 \le y < 25$	12	20
$25 \le y < 35$	4	30

Solution 39

a

Width $= 2(1.5) = 3$ cm. Area $8(1.5) = 12$ represents 24 plants. Thus frequency 12 needs area 6 and height 2 cm.

b

$Q_2 = 5 + (19/24)(5) = 8.96$ kg (or 9.06 with position 35.5).

c

$\bar{y} = 755/70 = 10.8$ kg.
$\sigma = \sqrt{12037.5/70 - (755/70)^2} = 7.46$ kg.

d

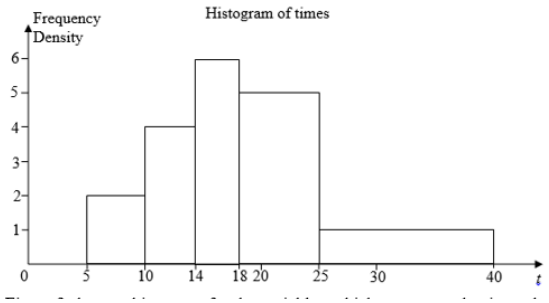
$\bar{y} > Q_2$, suggesting positive skew.

Supplementary Exercise

Q4. June 2007 Q5. The histogram shows times t (minutes) to swim 500 m.

(a) Complete the frequency table. (2 marks)
(b) Estimate how many took longer than 20 minutes. (2 marks)
(c) Estimate the mean. (4 marks)
(d) Estimate the standard deviation. (3 marks)
(e) Find the median and quartiles. (4 marks)
(f) Calculate $3(\text{mean} - \text{median})/\text{SD}$ and describe skewness. (2 marks)

Time	5–10	10–14	14–18	18–25	25–40
Frequency	10	16	24		



Solution 40

a

$7(5) = 35$ and $15(1) = 15$ complete the table.

b $(25 - 20)(5) + (40 - 25)(1) = 40$.

c

Midpoints 7.5, 12, 16, 21.5, 32.5. $\sum f = 100$, $\sum ft = 1891$. Mean $= 18.91$ min.

d

$\sigma = \sqrt{41033/100 - 18.91^2} = 7.26$ min.

e

$Q_2 = 18$. $Q_1 = 10 + (15/16)(4) = 13.75$. $Q_3 = 18 + (25/35)(7) = 23$.

f

$3(18.91 - 18)/7.26 = 0.376$, a positive skew.

Supplementary Exercise

Solution 41

Jan 2013 Q5. The table summarises weekly incomes of 100 households. Use $\sum fy^2 = 12452800$. The $200 \le y < 240$ histogram bar is 2 cm wide and 7 cm high.

(a) Find width and height for $320 \le y < 400$. (3 marks)

(b) Interpolate the median income to the nearest pound. (2 marks)

(c) Estimate the mean and standard deviation. (4 marks)

(d) Use $3(\text{mean} - \text{median})/\text{SD}$ to calculate and describe skewness. (2 marks)

Income (£)	Midpoint	Frequency
$0 \le y < 200$	100	12
$200 \le y < 240$	220	28
$240 \le y < 320$	280	22
$320 \le y < 400$	360	18
$400 \le y < 600$	500	12
$600 \le y < 800$	700	8

a

Width $= 4$ cm. Area 14 cm^2 represents 28 households. For 18 households area is 9, so height $= 9/4 = 2.25$ cm.

b

$240 + (10/22)(80) = 276.36$: £276.

c

$\sum fy = 31600$, so $\bar{y} = 316$.
$\sigma = \sqrt{12452800/100 - 316^2} = 157.07$ pounds.

d

$3(316 - 276)/157.07 \approx 0.76$, a positive skew.

Supplementary Exercise

Solution 42

May 2010 Q5. A random sample of 56 students recorded weekly television viewing to the nearest hour.

(a) Find the missing midpoints. (2 marks)

(b) The 11–20 bar measures 4 cm by 6 cm. Find the width and height for 26–30. (3 marks)

(c) Estimate the mean and SD. (5 marks)

(d) Interpolate the median. (2 marks)

(e) With $Q_1 = 15.8$ and $Q_3 = 29.3$, describe skewness and give a reason. (2 marks)

Hours	1–10	11–20	21–25	26–30	31–40	41–59
Frequency	6	15	11	13	8	3
Midpoint	5.5	15.5		28		50

a

23 and 35.5 hours.

b

Width $= 2$ cm. Height $= (13/5)(4) = 10.4$ cm.

c

$\sum fx = 1316.5$, $\sum fx^2 = 37378.25$.
$\bar{x} = 1316.5/56 = 23.5$ h and $\sigma = \sqrt{37378.25/56 - \bar{x}^2} = 10.7$ h.

d

$Q_2 = 20.5 + ((28 - 21)/11)(5) = 23.68$ h.

e

$Q_3 - Q_2 \approx 5.6 < 7.9 \approx Q_2 - Q_1$. Negative skew.

Supplementary Exercise

Solution 43

Jan 2009 Q5. 104 teenagers reported time spent shopping last month, to the nearest hour. The 8–10 histogram bar was 1.5 cm wide and 3 cm high.

(a) Find the width and height for 16–25 hours. (3 marks)

(b) Interpolate the median and IQR. (5 marks)

(c) Estimate the mean and SD. (4 marks)

(d) Describe skewness with a reason. (2 marks)

(e) Recommend an average and measure of spread with a reason. (2 marks)

Hours	Midpoint	Frequency
0–5	2.75	20
6–7	6.5	16
8–10	9	18
11–15	13	25
16–25	20.5	15
26–50	38	10

a

Widths 3 h and 10 h give width 5 cm. Densities $18/3 = 6$ and $15/10 = 1.5$ give height $3(1.5/6) = 0.75$ cm.

b

$Q_2 = 7.5 + (16/18)(3) = 10.2$. $Q_1 = 5.5 + (6/16)(2) = 6.25$. $Q_3 = 10.5 + (24/25)(5) = 15.3$. IQR approximately 9 h.

c

$\sum fx = 1333.5, \sum fx^2 = 27254$.
$\bar{x} = 1333.5/104 = 12.8, \sigma = \sqrt{27254/104 - \bar{x}^2} = 9.88$ h.

d

$Q_3 - Q_2 \approx 5.1 > 3.9 \approx Q_2 - Q_1$: positive skew.

e

Use median and IQR because they resist extreme values in skewed data.